

EDISON ALTAMIRANO

Robotics Engineer | VLSI Engineer | Hardware Engineer | Analog & RF Engineer

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<https://edisonaltamirano.github.io/myportfolio/>

LANGUAGES

Spanish Native ●●●●●

English Proficient ●●●●●

French Intermediate ●●●●●

SKILLS

Software

AUTOSAR, ROS1/2, C/C++, Python, Flutter, ReactJS, MATLAB, GCP/AWS

Hardware

PCB Design (KiCad-EasyEDA), HSPICE, LTspice, FMCW Radars, LiDAR (Ouster-Velodyne), RGB-D Cameras (ZED, Orbbec), STM32 / ESP32, JETSON Series NVIDIA

PUBLICATIONS

Visual Navigation Algorithms for Mobile Manipulators in Service Shops

11/2023
https://doi.org/10.1007/978-3-031-47640-2_19

Towards a Mixed Virtual Reality Environment Implementation to Enable Industrial Robot Programming Competencies within a Cyber-Physical Factory

05/2023
<https://doi.org/10.1109/EDUCON54358.2023.10125175>

A Digital Twin implementation for Mobile and collaborative robot scenarios for teaching robotics based on Robot Operating System

03/2022
<https://doi.org/10.1109/EDUCON52537.2022.9766583>

AWARDS

 **Most Consistent Solution in AIRLab Stacking Challenge**
Humanoid robot tournament hosted in the Netherlands, watch the project in action: [Link](#)

 **7 awards granted during RoboBoat, RoboSub 2020-2022**
Autonomous boat and submarine tournament hosted in the US: <https://robonation.org/>

CERTIFICATION

[Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization](#)

DeepLearning.AI (jul.2019)

EDUCATION



M.S. in Electrical Engineering

[Stanford University](#)

2025 - 2027

- Focus on Analog and Mixed-Signal Integrated Circuits, VLSI Systems, and Advanced Sensor Hardware/Software Integration



B.S. in Digital Systems and Robotics Engineering

[Instituto Tecnológico y de Estudios Superiores de Monterrey ITESM](#)

08/2018 - 06/2023

EXPERIENCE

Application Engineer

[ZF Group](#)

09/2023 - 09/2025 Monterrey, Mexico

- Engineered next-generation electronic braking ECUs, combining hardware validation, system architecture design, and model-based control.
- Built MATLAB/Simulink dynamic models and integrated AUTOSAR components using DaVinci and VectorCAST.
- Supported architecture reviews and implemented safety-critical diagnostics aligned with automotive standards.
- Improved system architecture and remote testing with CAPL scripting and CANalyzer.

CTO

[CHAKRI](#)

08/2020 - 09/2025 Quito, Ecuador

- Led and executed the development of a farm-to-home delivery platform, creating 3 mobile apps and a logistics website linking farmers and consumers. View the apps on Play Store or App Store: [Link](#)

ADAS Validation Engineer

[ZF Group](#)

08/2022 - 06/2023 Monterrey, Mexico

- Designed and validated STM32-based hardware modules for an autonomous shuttle (acceleration, braking, and lighting).
- Integrated CAN bus networks, synchronized LiDAR and radar data, and validated real-time performance with ROS2.
- Recognized by **top executives** from **BYD** and **ZF** during key demonstrations. Watch the project in action: [Link](#)

Research Assistant

[Tecnológico de Monterrey](#)

01/2019 - 08/2022 Monterrey, Mexico

- Built PLC interfaces and sensor boards for AGV fleet management.
- Linked reinforcement-learning control policies to hardware over Modbus/EtherCAT.
- Delivered robotics hardware integration training for 20+ faculty members.

LEADERSHIP AND PROJECTS

Graduate Researcher - Arbabian Lab

[Stanford University](#)

09/2025 - Present Stanford, United States

<https://arbabianlab.stanford.edu/>

- Conducting research in ambient sensing and multi-modal sensor fusion for autonomous and intelligent systems. Integrating TI IWR6843 70 GHz radar, Ouster LiDAR, and ZED, ELP, and Orbbec RGB-D cameras for synchronized perception pipelines.

Lead Software Engineer

[Vanttec](#)

08/2020 - 06/2023 Monterrey, Mexico

<https://vanttec.com/>

- Integrated control, vision, and navigation systems for virtual and real submarines in RoboSub and Roboat competitions from 2020 to 2022. Watch the project in action: [Link](#)

Software Developer

[RoBorregos](#)

10/2022 - 06/2023 Monterey, Mexico

<https://roborregos.com/>

- Developed an autonomous butler for the RoboCup @Home category in 2022-2023. Managed robot's SLAM and planning system. Watch the project in action at the IROS event in Japan: [Link](#)